

Mask Architecture Anomaly Segmentation for Road Scenes

Carmine Francesco Benvenuto
Politecnico di Torino
s333307@studenti.polito.it

Nicolò Colle
Politecnico di Torino
s336406@studenti.polito.it

Francesco Pio De Dominicis
Politecnico di Torino
francescopio.dedominicis@studenti.polito.it

Matteo Flores
Politecnico di Torino
matteo.flores@studenti.polito.it

github.com/matteo-flores/MaskArchitectureAnomaly_CourseProject

Abstract

*State-of-the-art semantic segmentation models often exhibit dangerous overconfidence when encountering out-of-distribution (OoD) objects. In this work, we address the challenge of **Road Anomaly Detection** by adapting the mask classification paradigm of Mask2Former (EoMT). We propose a robust fine-tuning strategy leveraging a dynamic **Outlier Exposure (OE)** pipeline to explicitly train the model on synthetic anomalies. We investigate distinct supervision objectives, comparing energy-based (RbA) and probabilistic (KL Divergence) formulations to enforce anomaly rejection. Extensive experiments on the **RoadAnomaly**, **RoadObstacle**, and **Fishyscapes** benchmarks demonstrate that our method significantly outperforms standard entropy-based baselines, effectively bridging the gap between closed-set segmentation and open-world reliability.*

1. Introduction

Standard semantic segmentation models operate under a **closed-world assumption**, often assigning high confidence to **Out-of-Distribution (OoD)** objects. This *overconfidence problem* [5] poses significant risks for safety-critical applications like autonomous driving. While recent approaches attempt to mitigate this via post-hoc uncertainty estimation or generative reconstruction, they often suffer from noise or prohibitive computational costs.

To bridge the gap towards **Open-World Semantic Segmentation**, we propose a robust anomaly detection framework built upon the **EoMT** (Mask2Former) architecture. We introduce a fine-tuning strategy leveraging dynamic

Outlier Exposure (OE), where synthetic anomalies are injected into training scenes on-the-fly. Crucially, we investigate two complementary supervision objectives to enforce anomaly rejection: the energy-based **Region-based Anomaly (RbA)** loss [10] and a probabilistic **KL Divergence** regularization [5], aimed at maximizing posterior entropy on outliers.

Our main contributions are summarized as follows:

- We propose a robust fine-tuning pipeline for Mask2Former architectures that effectively mitigates overconfidence via dynamic Outlier Exposure.
- We provide a comparative analysis of energy-based (RbA) and probabilistic (KL Divergence) objectives for mask-level anomaly detection.
- We demonstrate that efficient training strategies based on **Gradient Accumulation** enable high-resolution fine-tuning on resource-constrained hardware.
- We achieve competitive results on the **RoadAnomaly**, **RoadObstacle**, and **Fishyscapes** benchmarks, validating the effectiveness of our approach.

2. Related Work

Semantic Segmentation and Efficient Architectures. While Deep ConvNets are the standard for semantic segmentation, ERFNet [12] emerged as a premier real-time solution by employing factorized modules to minimize computational load. However, such models operate under a *closed-world assumption*, forcing pixels into fixed categories even when encountering Out-of-Distribution (OoD) objects. This leads to dangerous overconfidence in real-world scenarios. We use ERFNet as a representative pixel-based baseline to benchmark these limitations against mask-

based approaches.

Mask Classification Architectures. *Mask Classification* unifies segmentation tasks by predicting binary masks with global class labels. MaskFormer [2] established this paradigm, while Mask2Former [3] refined it via *Masked Attention*, significantly improving performance on small objects. We adopt this architecture because it naturally decouples mask generation from class prediction, offering a structural advantage in identifying anomalies that do not conform to any known category.

Self-Supervised Learning and Robust Features. Self-supervised learning, particularly DINOv2 [11], has revolutionized feature representation by learning robust, "object-centric" features without pixel-level annotations. By leveraging massive curated datasets, these foundation models inherently delineate semantic boundaries and offer superior generalization in open-world settings compared to standard supervised backbones.

Encoder-only Mask Transformer (EoMT). EoMT [7] combines the robustness of DINOv2 with the Mask2Former decoder. Adopting a "Frozen-ViT" strategy, it keeps the backbone frozen to preserve open-world generalization—avoiding overfitting to closed-set classes—while training only a lightweight decoder. This efficiency makes EoMT ideal for anomaly detection, as the frozen backbone retains a broad, label-agnostic understanding of objects.

Outlier Exposure (OE). To strictly enforce anomaly rejection, we integrate *Outlier Exposure* (OE) [5]. Unlike standard training, OE exposes the model to auxiliary synthetic anomalies (e.g., from COCO), minimizing a regularization loss (typically KL Divergence) to enforce a uniform posterior. This data-driven approach teaches the model heuristics to distinguish in-distribution samples from anomalies, significantly improving real-world generalization.

Anomaly Detection and Post-Hoc Methods. Traditional post-hoc methods like MSP, MaxLogit, and MaxEntropy operate on pixel-wise logits but often struggle with overconfidence or distinguishing near-distribution anomalies [6]. For mask architectures, Region-based Anomaly (RbA) [10] is preferred. Since object queries act as independent "one-vs-all" classifiers, RbA minimizes the "energy" of known classes in anomalous regions, defining an anomaly score based on the rejection of the region by all known queries.

3. Method

3.1. Anomaly Dataset for fine-tuning

Since standard urban semantic segmentation datasets, such as Cityscapes, contain exclusively in-distribution (ID) classes, we implemented an image-generation pipeline based on the Outlier Exposure (OE) paradigm to perform fine-tuning for the Anomaly Segmentation task. Unlike traditional static cut-and-paste approaches, our method adopts an on-the-fly strategy inspired by the Fishyscapes benchmark [1], enhanced with context-aware placement and photometric adaptation techniques to maximize the realism of the injected anomalies.

To simulate the presence of anomalous objects (out-of-distribution OOD), we construct a dataset¹ composed of images representing everyday objects and entities not included in the Cityscapes label space (e.g., toys, domestic and exotic animals) each accompanied by a binary-segmentation mask. During training, an anomalous object is randomly sampled and injected into a real-world scene drawn from the Cityscapes training set[4].

3.1.1. Pipeline

Data generation is performed dynamically during image loading: the custom `CityscapesAnomalyDataset` class applies the anomaly injection function with probability $p = 0.5$. This strategy ensures that, across training epochs, the model is exposed to an effectively unbounded set of object-background combinations, substantially reducing the risk of overfitting.

Domain Adaptation. However, directly pasting an object onto a different background typically introduces strong discontinuities at object boundaries as well as significant shifts in color distribution, which neural networks can easily exploit as shortcut cues for detection. To encourage the model to focus on semantic anomalies rather than on low-level artifacts introduced during the initial composition stage, we implemented a set of harmonization techniques:

- **Color Transfer:** the color statistics (mean and standard deviation) of the anomalous object are adapted to match those of the background image.
- **Depth-dependent Blur:** we simulate depth-of-field effects by applying a variable Gaussian blur: objects located near the horizon (central region of the image) or extremely close to the camera (lower region) undergo stronger smoothing.
- **Noise Injection:** additive noise is applied to the object to mimic the sensor grain characteristic of Cityscapes images.

¹The dataset is available at <https://drive.google.com/uc?id=1ANULaCsFV3aumHi-xaDB8gFSdEE4K86Y>



Figure 1. Visual comparison of the anomaly injection pipeline. **Left:** The original in-distribution scene from Cityscapes. **Center:** *Naive injection*, where the anomalous object is pasted directly; **Right:** *Harmonized injection* using our proposed Domain Adaptation techniques.

- **Edge Smoothing:** the object is spatially smoothed to avoid sharp transitions between anomaly and background.

Figure 1 represents all the techniques previously mentioned.

Context-Aware Placement. A purely random placement of anomalous objects may result in physically impossible configurations (e.g., a floating dog when placed in the upper region of the image) or occlude critical scene elements such as the ego-vehicle. To mitigate these issues, we design a constraint-aware object placement algorithm, which enforces the following rules:

- **Label-based constraint:** each anomalous object is assigned to a label: air, ground, or generic object (`obj`). According to this label, the object is preferentially placed within specific regions of the background.
- **Ego-Vehicle Avoidance:** if the injected anomaly overlaps with the ego-vehicle region beyond a predefined tolerance threshold, the candidate location is discarded and resampled.

Data Augmentation. Finally, we apply a set of standard data augmentation techniques to artificially increase both the size of the training set and the diversity of the generated samples. Specifically, we employ:

- **Random Flip**
- **Random Rotation**
- **Color Jitter**
- **Saturation Jitter**

Output Representation. The final output of the pipeline consists of an image I_{final} , obtained by composing a Cityscapes background with an injected anomalous object, and an updated segmentation mask M_{final} , in which pixels corresponding to the anomaly are assigned to a dedicated class (`anomaly_id`).

3.2. RbA Loss

To address the challenge of anomaly segmentation in open-world settings, we adopted and adapted the **Region-based Anomaly (RbA)** methodology recently proposed by Nayal *et al.* [10]. This approach is distinguished by defining the

“anomaly” event not as a separate class but as a complementary event defined by being “rejected by all known classes”.

3.2.1. Theoretical Formulation and Adaptation

The Mask2Former architecture [3] (and consequently EoMT [7]) operates through a mask classification paradigm, decoupling spatial mask prediction from semantic classification. Nayal *et al.* demonstrate that object queries in these architectures tend to behave as independent “one-vs-all” classifiers. Leveraging this property, we implemented the RbA Loss to penalize the activation of known classes in regions identified as **Out-of-Distribution (OoD)**. In the forward method of our RbALoss class, the calculation proceeds through the following steps:

1. **Pixel-wise Reconstruction:** Since the model predictions are region-based (N queries), we aggregate probabilities to obtain a dense membership map for known classes (K). We define the probability that pixel i belongs to the known distribution P_{inlier} as the weighted sum of mask probabilities P_M and class probabilities P_C :

$$P_{\text{inlier}}(i) = \sum_{q=1}^N P_M(q, i) \cdot P_C(q, c_{\text{known}}) \quad (1)$$

This operation is efficiently implemented via Einstein summation (`torch.einsum`).

2. **Energy Minimization:** In accordance with Nayal *et al.*’s formulation [10], the objective is to minimize the probability that an anomalous pixel belongs to any of the known classes. For pixels belonging to the anomaly mask Ω_{out} , we define the loss function as a variant of the squared hinge loss proposed in the original paper:

$$\mathcal{L}_{\text{RbA}} = \frac{1}{|\Omega_{\text{out}}|} \sum_{i \in \Omega_{\text{out}}} (P_{\text{inlier}}(i))^2 \quad (2)$$

Adopting a squared penalty forces the model to drastically suppress known activations in ambiguous regions, consequently maximizing the anomaly score $\text{RbA}(x)$, defined as the complement of known probabilities.

3.2.2. Fine-Tuning Strategy and Outlier Exposure

Following the observations of Nayal *et al.*, direct training on outliers can degrade performance on known classes (closed-set performance) if not managed correctly. To mitigate this risk, we implemented a Minimal Supervision Fine-Tuning strategy. Our training pipeline adopts a hybrid approach:

1. **Parameter Freezing:** As suggested in the literature to preserve learned robust features, we freeze the backbone and pixel decoder, optimizing exclusively the prediction heads and learned queries. This is handled by the `get_optimizer` function, which selects only parameters with `requires_grad=True`.

2. Dual-Branch Training: The training loop dynamically distinguishes between “clean” samples and samples containing anomalies (outlier exposure):

- **For Clean images,** we apply the standard Mask2Former loss (MaskClassificationLoss), ensuring the model retains the ability to segment Cityscapes classes.
- **For OoD images** (identified by `ANOMALY_ID`), the flow is diverted to apply exclusively the $\mathcal{L}_{\text{RbA}}$ scaled by a factor $\lambda = 0.5$.

This logical separation prevents the model from “hallucinating” known classes on unknown objects, resolving the overconfidence problem typical of softmax classifiers, as highlighted in [6, 10]. Experimental results confirm that this approach improves separability between inliers and outliers, reducing false positives at the boundaries of known classes.

3.3. KL Divergence Loss

As an alternative to the energy-based formulation of RbA, we explored a probabilistic approach rooted in the **Outlier Exposure (OE)** framework introduced by Hendrycks *et al.* [5]. While standard cross-entropy encourages low entropy (high confidence) on in-distribution data, it provides no explicit supervision for the anomaly space. This often leads to the *overconfidence problem*, where the model assigns high probability to known classes even when encountering semantically unknown objects [6].

To mitigate this, we incorporate a regularization term based on the **Kullback-Leibler (KL) Divergence**. The core intuition is to enforce maximum entropy on the model’s posterior distribution when processing anomalous pixels. Mathematically, for an input x belonging to the outlier distribution $\mathcal{D}_{\text{out}}$, we want the predicted class distribution $f(x)$ to approximate a Uniform distribution $\mathcal{U}$.

3.3.1. Formulation

In the context of the Mask2Former architecture, we apply this constraint to the classification logits of the queries associated with anomalous regions. The auxiliary loss component is defined as:

$$\mathcal{L}_{\text{OE}} = \mathbb{E}_{x \sim \mathcal{D}_{\text{out}}} [D_{\text{KL}}(\mathcal{U} || f(x))] \quad (3)$$

Minimizing this divergence penalizes any peak in the probability distribution over the K known classes. Effectively, this forces the model to say “I don’t know” rather than hallucinating a specific class label. The final training objective becomes a weighted sum of the standard mask classification loss ($\mathcal{L}_{\text{std}}$) for clean data and the outlier exposure term:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{std}} + \lambda_{\text{oe}} \mathcal{L}_{\text{OE}} \quad (4)$$

where λ_{oe} balances the trade-off between segmentation accuracy and anomaly uncertainty. Empirically, we found that this explicit uncertainty maximization complements the discriminative power of the mask queries, leading to smoother decision boundaries around known manifolds.

3.4. Optimization Strategy

Training Transformer-based architectures like EoMT [7] on high-resolution urban imagery (1024×2048 px) poses significant hardware constraints. Specifically, the memory footprint required to store attention maps and activations exceeds the capacity of standard commodity GPUs (e.g., NVIDIA T4, 16GB VRAM), forcing the adoption of a unitary physical batch size ($B = 1$).

However, training with a batch size of 1 introduces significant noise in the gradient estimation, leading to unstable convergence and poor generalization. To address this engineering bottleneck without sacrificing input resolution, we implemented a **Gradient Accumulation** strategy.

Instead of updating the model weights at every iteration, we accumulate gradients over N_{acc} forward-backward steps. The optimizer step is performed only after processing an effective batch of size $B_{\text{eff}} = B \times N_{\text{acc}}$. This approach allows us to:

1. **Stabilize Training:** The accumulated gradient provides a more accurate estimate of the true loss landscape direction, smoothing out the noise introduced by individual high-variance samples (especially those with heavy CutPaste augmentations [1]).
2. **Simulate Large Batches:** We simulate an effective batch size of 16 (using $N_{\text{acc}} = 16$), which is crucial for the stability of the AdamW optimizer [9] and for proper statistics in normalization layers.

Formally, if g_t is the gradient at step t , the accumulated update $\Delta\theta$ is computed as $\Delta\theta \propto \sum_{i=0}^{N_{\text{acc}}-1} g_{t-i}$. This technique proved instrumental in stabilizing the training process and achieving competitive performance within limited computational resources.

3.5. ODIN in EoMT

While we employ fine-tuning strategies to adapt the model to the anomaly segmentation task, we also investigate ODIN [8] as a training-free, post-hoc mechanism to improve OOD detection. Designed to amplify the separability between ID and OOD samples without modifying model parameters, ODIN operates directly during inference.

The method was originally proposed in the context of image classification, where a neural network produces, for each input image, a vector of logits $f(x) \in \mathbb{R}^K$, with K known classes.

The key observation behind ODIN is that OOD samples tend to produce flatter probability distributions, whereas ID samples yield more confident predictions. To amplify

this separation, ODIN introduces two fundamental modifications to the inference process:

1. **Temperature Scaling of logits**, to reduce the saturation of the softmax function and make the differences between in-distribution (ID) and out-of-distribution (OOD) samples more evident.
2. **Image Perturbation**: A small, input-dependent noise is added to the input image. This perturbation aims to *maximize* the model’s confidence in its own prediction. By slightly shifting the input pixels in the direction that increases the predicted class probability, the score for ID images is significantly enhanced, while producing significantly weaker effects on OOD inputs, widening the separability gap.

3.5.1. Strategies

The application of ODIN, originally designed for standard CNN-based classifiers, to the EoMT architecture [7] poses significant structural challenges.

EoMT is a mask-based transformer model that operates through object queries. More specifically, EoMT does not produce a single global prediction per image, but it generates a decomposition based on multiple object queries. Specifically, the output of the model consists of:

- **Classification logits per query**: $L_q \in \mathbb{R}^{K+1}$
- **Mask logits**: $M_q \in \mathbb{R}^{H \times W}$

where q denotes a learnable object query.

To extend ODIN to this setting, we propose and implement two distinct strategies for computing the gradient and the corresponding input perturbation.

Pixel-wise Strategy. EoMT does not directly produce a pixel-wise probability map, but instead outputs a set of Q class-mask pairs $\{(L_q, M_q)\}_{q=1}^Q$, which can be combined to construct the final semantic probability map through a semantic fusion process:

$$P(c | x_{ij}) = \sum_q \sigma(M_q(i, j)) \cdot \text{softmax}(L_q), \quad P \in \mathbb{R}^{K \times H \times W} \quad (5)$$

For each pixel (i, j) in the image, a probability vector over the semantic classes is therefore available:

$$\mathbf{p}_{ij} = [P(1 | x_{ij}), \dots, P(K | x_{ij})] \in \mathbb{R}^K \quad (6)$$

The core idea of this strategy is to interpret each pixel as an independent classification problem. Consequently, similarly to the original formulation of ODIN [8], we define the confidence associated with pixel (i, j) as follows:

$$s_{ij} = \max_c P(c | x_{ij}) \quad (7)$$

where s_{ij} is a scalar representing the confidence of the model in the predicted class for that pixel. Consequently,

the local pixel-wise loss is defined as:

$$\ell_{ij} = -\log(s_{ij}) = -\log\left(\max_c P(c | x_{ij})\right) \quad (8)$$

The overall image-level loss is given by:

$$\mathcal{L}_{\text{pixel}} = \sum_{i,j} \ell_{ij} \quad (9)$$

Query-level Strategy. Recognizing that the fundamental computational unit of EoMT is the object query rather than the individual pixel, we develop a second strategy that operates directly in the query space.

In the EoMT architecture [7], the model produces two distinct outputs: $L_q \in \mathbb{R}^{K+1}$, which expresses the probability distribution over the semantic categories, and $M_q \in \mathbb{R}^{H \times W}$, which delineates its spatial extent within the image.

In essence, while L_q answers the question “*which class does this query represent?*”, M_q specifies where this object is active in the image. Consequently, the semantic information regarding the object category is entirely contained in the classification logits, whereas the mask only intervenes at a later stage to project such information into the pixel space.

Based on this observation, we identify the classification logits of the object queries as the most natural location at which to define the notion of confidence required by ODIN. By applying temperature scaling to the logits of each query,

$$\tilde{L}_q = \frac{L_q}{T}, \quad (10)$$

a probability distribution over the known classes can be obtained via the softmax function:

$$p_q = \text{softmax}(\tilde{L}_q) \quad (11)$$

Following the original ODIN principle, we define the confidence of a query as the maximum probability associated with its class distribution:

$$s_q = \max_k p_{q,k} \quad (12)$$

However, in the EoMT architecture, the number of object queries Q is fixed a priori and is generally larger than the number of semantic entities actually present in the scene. As a consequence, only a fraction of the queries correspond to semantically meaningful objects, while the remaining ones are assigned to the void class or remain unused. Applying the ODIN objective indiscriminately to all queries would therefore be inappropriate: queries predicting the void class tend to exhibit very high confidence values, which would dominate the loss function and induce an input perturbation oriented toward suppressing semantic

Model	Method	mIoU	SMIYC RA-21		SMIYC RO-21		FS L&F		FS Static		Road Anomaly	
			AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95
ERFNET	MSP	72.20%	29.08%	62.63%	2.71%	65.27%	1.75%	50.62%	7.43%	41.84%	12.43%	82.60%
	MaxLogit		38.29%	59.43%	4.63%	48.48%	3.30%	45.52%	9.44%	40.32%	15.55%	73.31%
	MaxEntropy		30.95%	62.74%	3.04%	65.96%	2.58%	50.19%	8.77%	41.58%	12.67%	82.75%
EoMT	MSP	81.48%	69.43%	14.31%	85.95%	3.76%	16.69%	13.31%	55.68%	44.71%	69.42%	23.85%
	MaxLogit		69.18%	14.09%	85.86%	3.79%	16.69%	13.08%	55.91%	48.84%	68.98%	23.52%
	MaxEntropy		69.91%	13.84%	85.83%	3.78%	18.65%	13.12%	54.71%	44.94%	71.46%	23.08%
	RbA		68.73%	13.96%	76.77%	3.79%	10.81%	12.99%	39.73%	45.38%	54.97%	23.40%

Table 1. Anomaly detection for ERFNET and EoMT models. The masked-based model outperform better than the pixel-based one, on all the proposed datasets and for all the methods used for classify anomalies.

Model	Method	mIoU	SMIYC RA-21		SMIYC RO-21		FS L&F		FS Static		Road Anomaly	
			AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95
ERFNET	MSP ($t = 0.5$)	72.2%	27.06%	62.78%	2.42%	63.44%	1.28%	62.87%	6.60%	43.61%	12.18%	82.26%
	MSP ($t = 0.75$)		28.16%	62.49%	2.57%	64.16%	1.49%	51.74%	6.99%	42.50%	12.32%	82.32%
	MSP ($t = 1.1$)		29.40%	62.65%	2.76%	65.87%	1.86%	50.18%	7.69%	41.62%	12.46%	82.73%
	MSP (Best t)		30.64%	66.70%	3.00%	72.67%	3.40%	42.86%	11.60%	61.26%	12.66%	85.56%
	<i>Optimal t value</i>		<i>($t=2.5$)</i>		<i>($t=2.0$)</i>		<i>($t=100$)</i>		<i>($t=100$)</i>		<i>($t=2.5$)</i>	
EoMT	MSP ($t = 0.5$)	81.5%	66.81%	34.20%	88.20%	4.16%	28.09%	7.71%	55.33%	41.54%	68.26%	21.62%
	MSP ($t = 0.75$)		68.30%	18.92%	87.15%	3.90%	31.98%	5.69%	56.62%	41.30%	69.78%	23.01%
	MSP ($t = 1.1$)		69.68%	13.20%	85.67%	3.88%	33.38%	6.07%	58.38%	47.20%	70.20%	24.53%
	MSP (Best t)		72.89%	30.47%	89.11%	1.26%	36.33%	7.85%	59.99%	67.75%	70.20%	24.53%
	<i>Optimal t value</i>		<i>($t=10.0$)</i>		<i>($t=0.2$)</i>		<i>($t=100$)</i>		<i>($t=2.0$)</i>		<i>($t=1.1$)</i>	

Table 2. Temperature Scaling results for both models. We report performance for fixed temperatures and for the optimal t (indicated in italics below the best score) found via grid search.

content, rather than strengthening the separation between known classes and anomalies.

To prevent this behavior, the ODIN objective is restricted to a subset of semantically meaningful queries. Specifically, we define the set of valid queries as:

$$\mathcal{Q}_{\text{valid}} = \{q \mid \arg \max_k p_{q,k} \neq k_{\text{void}} \wedge \max_k p_{q,k} > \tau\}, \quad (13)$$

where k_{void} denotes the index of the void class and τ represents a minimum confidence threshold. The query-level ODIN loss is then defined as:

$$\mathcal{L}_{\text{ODIN-query}} = \sum_{q \in \mathcal{Q}_{\text{valid}}} \log(\max_k p_{q,k}). \quad (14)$$

4. Experiments

4.1. ERFNET vs EoMT

As a preliminary analysis, we evaluate the performance of the standard implementations of the ERFNET and EoMT models, pretrained on Cityscapes images.

Initially, the mIoU value has been computed for both models, to verify that the segmentation performance is consistent with the state of the art on the training set on which they

have been trained.

To evaluate which pixel is classified as anomaly and which not, 3 methods have been used [6]:

- **MSP** (Maximum Softmax Probability), which computes, as its acronym suggests:

$$\text{MSP}(x) = \max_c P(y = c \mid x) = \max_c \frac{e^{z_c(x)}}{\sum_{k=1}^K e^{z_k(x)}} \quad (15)$$

If the probability of the predicted class 15 is not sufficiently high, this means that the selected pixel is an anomaly. This method is likely to be overconfident, since the *softmax* function normalizes the logits into a probability distribution that sums to 1, often masking the true confidence.

- **MaxLogit**, which uses a similar function to MSP 15, but avoids the softmax and acts directly on class logits:

$$\text{MaxLogit}(x) = \max_c z_c(x) \quad (16)$$

Unlike MSP, the compared values are not previously *stretched*, and this allows to preserve their magnitude. For example, if the logits are all approximately 0, *softmax* may still return a high class probability.

Loss	Method	mIoU	SMIYC RA-21		SMIYC RO-21		FS L&F		FS Static		Road Anomaly	
			AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95
KL diverg.	MSP	81.13%	↑74.47%	↑14.94%	↓79.89%	↓0.95%	↑19.57%	↓8.20%	↑75.23%	↓8.27%	↑72.10%	↓15.72%
	MaxLogit		↑75.42%	↑14.76%	↑87.75%	↓0.95%	↑20.81%	↓8.03%	↑79.13%	↓7.97%	↑75.19%	↓15.58%
	MaxEntropy		↑72.87%	↑15.18%	↓60.03%	↓0.95%	↓15.98%	↓8.08%	↑61.37%	↓7.71%	↓67.60%	↓15.48%
	RbA		76.95%	14.53%	86.72%	0.95%	15.56%	7.30%	74.69%	6.68%	66.76%	15.22%
RbA loss	MSP	81.23%	71.12%	18.62%	78.26%	0.93%	18.96%	7.89%	72.87%	8.96%	71.51%	12.47%
	MaxLogit		70.59%	18.49%	87.70%	0.94%	21.49%	7.75%	78.95%	8.71%	72.06%	12.35%
	MaxEntropy		69.88%	19.26%	62.05%	0.94%	15.61%	7.68%	59.48%	8.47%	64.90%	12.26%
	RbA		↑74.45%	↑18.38%	↑88.04%	↓0.92%	↑16.85%	↓7.18%	↑75.50%	↓7.49%	↑68.11%	↓12.18%

Table 3. Quantitative results comparing Baseline and Outlier Exposure (OE) fine-tuning. The arrow highlight the values we have to focus on, the other are reported for completeness. Green arrows indicate improvement, red arrows indicate worsening. Bold arrows indicate strong improvement/worsening, i.e. more than 10%.

- **MaxEntropy**, which instead measures the uncertainty of the probability distribution, following the Entropy of Shannon:

$$H(x) = - \sum_{c=1}^K P(y = c | x) \cdot \log P(y = c | x) \quad (17)$$

The higher the entropy, the higher the confidence that we are dealing with an anomaly.

Additionally, **RbA** method has been used, applied on the masked-based model.

The goodness of anomaly scores have been evaluated using Area Under Precision-Recall Curve and False Positive Rate at 95% recall.

As expected (Tab. 1), pixel-based models (ERFNET, in our case) perform better in finding big objects as anomalies (see RoadAnomaly21 statistics), but results on tiny objects are catastrophic (see FS Lost&Found, RoadObstacle21 and Road Anomaly).

EoMT shows improvements with respect to ERFNET: considering AuPRC, EoMT improved by 40% – 60% on the big-anomalies dataset and up to 80% on the small-objects dataset. This proves that masked-based models are intrinsically more robust, able to better identify small objects. Nevertheless, RbA scores performs worse than all the other 3 anomaly score methods using EoMT. This lack will be solved with appropriate finetuning (Sec. 4.3).

4.2. Temperature scaling

In order to have another baseline to compare our fine-tuned models to, class logits have been rescaled through the temperature scaling technique, obtaining the best anomaly scores for anomaly segmentation. It has been applied on both ERFNET and EoMT (Tab. 2), on predefined temperatures (0.5, 0.75, 1.1) and on the (approximately) best temperature for each dataset, for each model.

4.3. Outlier Exposure

Focusing on improving EomT, as explained in Sec. 3, the two heads have been trained, using 2 losses:

- **KL divergence**, to directly improve MSP, MaxLogit and MaxEntropy score.
- **RbA loss**, to directly improve RbA .

The fine-tuning process was initially conducted, for both methods, using 4-sized batches, on Google Colab environment, due to hardware constraints. The batch size has been chosen according to the training setting of the mask-based models [3]. However, preliminary results indicated that these fine-tuned models performed worse than the original EoMT. We attribute this degradation to the instability of Batch Normalization (BN) statistics, since BN can warp mean and variance of images without anomalies, since each batch may contain "clean" in-distribution images and anomaly-injected samples (generated via the *cut-paste* strategy described in Sec. 3).

In order to bypass the "BN bug", we used 1-sized batches and in conjunction with the gradient accumulation technique which decouples the forward pass from the optimizer step, simulating bigger batches for weights update.

For both losses, the model has been trained using 4 epochs and saving parameters at each epoch. At the end, the 4-epoch trained (KL divergence) and the 2-epoch trained models (RbA loss) peaked, and their performance are reported in Tab. 3.

For a first overview, it is immediately clear that KL divergence boosts not only MSP, MaxLogit, and MaxEntropy scores, but also RbA score, and vice-versa. Furthermore, the performance difference between the two scores is negligible (the maximum difference is about 3%).

Our results using Outlier Exposure clearly show that there is a quite large improvement in almost all methods and for all analyzed datasets. The score that benefits the most is RbA, which performed slightly worse than the other 3 methods before the fine-tuning (Sec. 4.1) and now has similar per-

Model	Method	mIoU	SMIYC RA-21		SMIYC RO-21		FS L&F		FS Static		Road Anomaly	
			AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95
EoMT	MSP	81.48%	69.72%	14.36%	85.65%	3.76%	18.74%	13.11%	54.23%	52.21%	70.41%	23.94%
	Pixel-wise ODIN		18.99%	99.16%	29.61%	99.99%	24.09%	97.85%	28.10%	99.66%	30.49%	98.47%
	Query-level ODIN		53.96%	98.80%	50.57%	99.99%	20.83%	59.98%	54.20%	99.28%	63.91%	90.78%

Table 4. Anomaly detection comparison on EoMT. The table reports results for standard MSP and the proposed Pixel-wise ODIN and Query-level ODIN across multiple benchmarks.

formance. The “cost” of this improvement is the loss of just **0.35%** and **0.25%** of the mIoU on Cityscapes, which proves that the model can still perform well, despite the finetuning.

4.4. ODIN in EoMT

Pixel-wise strategy Even if the formulation described in eq.6 represents a natural extension of the ODIN principle to the semantic segmentation setting, this approach does not lead to significant improvements in anomaly detection performance when applied to the EoMT architecture. As shown in Table 4, the Pixel-wise strategy fails to produce valid results, reinforcing the limitations of this approach.

We attribute this behavior to the following reason. In its original formulation, ODIN operates on a single global confidence measure, which yields a unique scalar objective function. As a result, the loss gradient is oriented along a single direction, corresponding to an increase in the confidence of the image-level prediction.

Under the proposed pixel-wise strategy, this assumption no longer holds. The confidence is no longer represented by a single scalar summarizing the entire image, but by thousands of local confidence values s_{ij} that are potentially misaligned with each other.

Query-level strategy Experimental results, in Table 4, show that adapting ODIN at the object-query level yields superior performance compared to the pixel-wise formulation described in previously. However, despite the observed improvement, this strategy remains in general unsatisfactory for anomaly detection, highlighting structural limitations related both to the operating principle of ODIN and to the nature of mask-based architectures.

In our query-level formulation, we exclusively optimize the classification logits L_q , implicitly assuming that higher semantic confidence leads to improved ID and OOD separation. Nevertheless, the final anomaly map inevitably depends on the mask logits M_q , which are required to project the anomaly score into the pixel space. Since the ODIN gradient is computed modifying only the input in order to maximize L_q , there is no guarantee that the corresponding mask M_q remains coherent.

This behavior clearly emerges from the qualitative analysis reported in Figure 2. In particular, the perturbation gen-

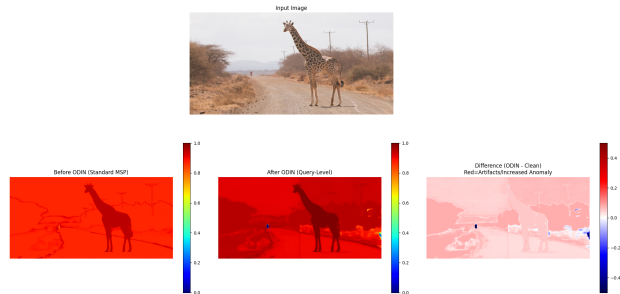


Figure 2. Qualitative analysis of ODIN perturbation on EoMT.

erated by ODIN induces variations along object boundaries, as highlighted in the amplified noise map. When comparing the anomaly map obtained using standard MSP with the one produced after applying query-level ODIN, it can be observed that the perturbation artificially increases the anomaly score in the vicinity of object edges. This effect is further emphasized in the difference map, where red regions indicate an increase in the anomaly value with respect to the clean prediction.

5. Conclusion

In this work, we addressed the critical challenge of overconfidence in semantic segmentation for autonomous driving by adapting the Encoder-only Mask Transformer (EoMT) architecture for Road Anomaly Detection. We demonstrated that mask-classification paradigms inherently outperform standard pixel-based baselines (e.g., ERFNet), particularly in identifying small-scale obstacles that pose significant safety risks.

Our proposed fine-tuning strategy, leveraging dynamic Outlier Exposure and a hardware-efficient Gradient Accumulation pipeline, effectively bridges the gap between closed-set accuracy and open-world reliability. Experimental results on the RoadAnomaly, RoadObstacle, and Fishyscapes benchmarks confirm that explicitly enforcing anomaly rejection (via both energy-based (RbA) and probabilistic (KL Divergence) objectives) yields superior performance compared to post-hoc methods like ODIN, which we found ill-suited for query-based architectures.

References

- [1] Hermann Blum, Paul-Edouard Sarlin, Juan Nieto, Roland Siegwart, and Cesar Cadena. The fishyscapes benchmark: Measuring blind spots in semantic segmentation. *Int. J. Comput. Vis.*, 129(11):3119–3135, 2021. [2](#), [4](#)
- [2] Bowen Cheng, Alexander Schwing, and Alexander Kirillov. Per-pixel classification is not all you need for semantic segmentation. In *Adv. Neural Inform. Process. Syst.*, 2021. [2](#)
- [3] Bowen Cheng, Ishan Misra, Alexander G Schwing, Alexander Kirillov, and Rohit Girdhar. Masked-attention mask transformer for universal image segmentation. In *IEEE Conf. Comput. Vis. Pattern Recog.*, pages 1290–1299, 2022. [2](#), [3](#), [7](#)
- [4] Marius Cordts, Mohamed Omran, Sebastian Ramos, Timo Rehfeld, Markus Enzweiler, Rodrigo Benenson, Uwe Franke, Stefan Roth, and Bernt Schiele. The cityscapes dataset for semantic urban scene understanding. In *IEEE Conf. Comput. Vis. Pattern Recog.*, 2016. [2](#)
- [5] Dan Hendrycks, Mantas Mazeika, and Thomas Dietterich. Deep anomaly detection with outlier exposure. In *Int. Conf. Learn. Represent.*, 2019. [1](#), [2](#), [4](#)
- [6] Dan Hendrycks, Steven Basart, Mantas Mazeika, Andy Zou, Joseph Kwon, Mohammadreza Mostajabi, Jacob Steinhardt, and Dawn Song. Scaling out-of-distribution detection for real-world settings. In *International Conference on Machine Learning (ICML)*, 2022. [2](#), [4](#), [6](#)
- [7] Tommie Kerssies, Niccolò Cavagnero, Giuseppe Averta, Bastian Leibe, Narges Norouzi, Alexander Hermans, Gijs Dubbelman, and Daan de Geus. Your vit is secretly an image segmentation model. In *IEEE Conf. Comput. Vis. Pattern Recog.*, 2025. [2](#), [3](#), [4](#), [5](#)
- [8] Shiyu Liang, Yixuan Li, and R. Srikant. Enhancing the reliability of out-of-distribution image detection in neural networks, 2020. [4](#), [5](#)
- [9] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *Int. Conf. Learn. Represent.*, 2019. [4](#)
- [10] Promit Nayal, K. Javed, and M. Shah. Rba: Region-based anomaly detection. In *IEEE Conf. Comput. Vis. Pattern Recog.*, 2023. [1](#), [2](#), [3](#), [4](#)
- [11] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy V Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, et al. Dinov2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024. [2](#)
- [12] Eduardo Romera, José M Álvarez, Luis M Bergasa, and Roberto Arroyo. Erfnet: Efficient residual factorized convnet for real-time semantic segmentation. *IEEE Trans. Intell. Transp. Syst.*, 19(1):263–272, 2018. [1](#)